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Predicting Mussel Motion for Autonomous ROV Mussel Sampling

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Abstract

Context: The University of Canterbury Computer Vision Department has developed a stereo-camera-equipped remotely operated vehicle (ROV) for collecting mussel samples on dropper lines in mussel farms. This aims to replace costly manual sampling. Reliable automatic alignment and grasping of the mussels requires anticipating the position of a mussel as it drifts with local currents, but the complexity of these currents makes this difficult to model directly.

Objective: This research investigates whether the ROV's own stereo-tracked mussel position data can be used to build a lightweight, real-time motion prediction model, sufficiently accurate and computationally efficient for onboard deployment, that consistently outperforms a zero-order-hold (ZOH) baseline.

Method: Three candidate prediction techniques frequency analysis and extrapolation, linear and polynomial interpolation, and an Unscented Kalman Filter (UKF) with a constant-velocity motion model. These methods were implemented in Python and benchmarked against ZOH across four real motion-capture datasets, using root-mean-square error (RMSE) at a 5s prediction horizon as the primary metric.

Findings: The UKF was the most accurate and stable of the three methods, achieving an average RMSE of 0.086 m versus 0.305 m (FFT) and 0.147–0.566 m (interpolation), and was the only method to improve, on average, over the ZOH baseline (+4.6%). This improvement was not consistent across all four datasets, and known contamination of RMSE by tracking dropouts recorded as (0,0,0) means the true performance gap may be understated. The UKF has therefore been selected as the primary prediction mechanism for further development and refinement.

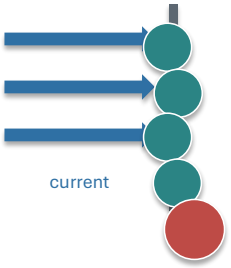
Graphical abstract

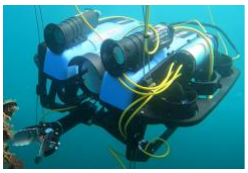
Predicting Mussel Motion for Autonomous ROV Mussel Sampling

Stereo-camera tracking → prediction → improved control response to grasp mussels

THE PROBLEM

Ocean currents and mussel dropper lines make it difficult for the ROV to grasp the mussels without overshooting their position as they drift.





ROV + stereo cameras

Research Workstreams

1 TRACK

Stereo cameras track mussel position through time



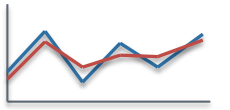
2 PREDICT

Candidate methods

- FFT
- 1st-order interpolation
- 2nd-order interpolation
- UKF

3 EVALUATE

Compare predicted vs observed



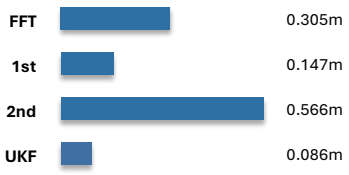
KEY RESULT

Unscented Kalman Filter (UKF) selected

0.086 m

Average RMSE

Average RMSE across 4 tests



APPLICATION

Use the UKF prediction as a lightweight estimate of target motion, providing the control system with forward-looking information to help the ROV remain aligned with the moving mussel line.

Outcome: more stable tracking and sampling in dynamic ocean conditions

1 Introduction

1.1 Context

The University of Canterbury (UC) Computer Vision Department has been working closely with mussel farms to develop an automated remotely operated vehicle (ROV), see Figure 1, capable of retrieving a sample of mussels from mussel farm drop lines before harvest. This is currently performed using a crane mounted on a large boat to lift the drop lines out of the water. This is costly to run and takes considerable man-hours to retrieve the necessary samples. The ROV would allow a much smaller boat and fewer staff to be used to collect the samples, drastically reducing costs for the mussel farmers. Thus, this has been a high-interest project for these mussel farmers who would benefit from the reduction in expenses throughout their operation.

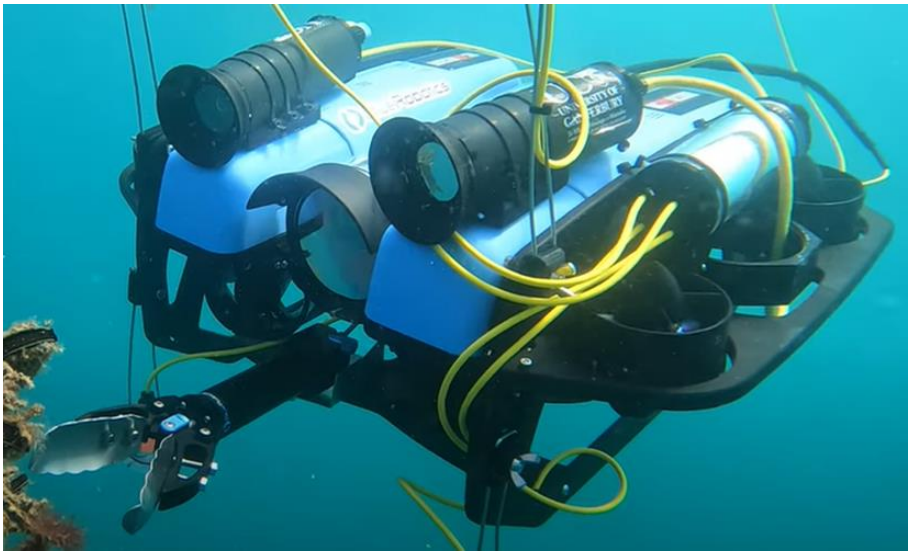


Figure 1: UC Vision Mussel Rope Scanning ROV. From: https://ucvision.org.nz/aquaculture/

1.2 Area Under Examination

Presently, the ROV is functional. It can navigate in the ocean in all directions, identifying mussels on the drop lines, and can grab mussels under manual operation. However, for the ROV to be a feasible replacement for the current crane system, it needs to be able to automatically align and grab the mussels off the line [1], as manual alignment is highly tedious and can take considerable time to retrieve just one mussel.

This research project initially aimed to investigate modelling the fluid motion in ocean/water-based structures, such as mussel farms. Specific investigations will be conducted to determine contributing factors and work towards developing the final model. These investigations will look into pre-existing models, how the structures impact fluid dynamics, and how measurable factors can be used to develop a controllable model.

As this research has progressed, the specific approach, and subsequently the scope of this research, has narrowed and shifted. Instead of developing a hydrodynamic model of a mussel farm, the focus has been moved to developing a robust prediction algorithm to estimate the position of mussels tracked by the ROV's stereo cameras and computer vision algorithm. This change aligns the research far more closely with the project and allows for more meaningful data and analysis to be conducted.

2 Literature

2.1 Mussel Farms and Infrastructure

Mussel farming is a major component of international seafood operations and is an environmentally stable form of seafood farming. The New Zealand mussel market largely focuses on the production of the green-lipped mussel. Production of these is largely based in regions such as the Marlborough Sounds, Tasman Bay, and the Coromandel Peninsula, as these locations have the right conditions to sustain consistent shellfish growth. Mussel farming is most practised using a suspended longline system where multiple pre-seeded dropper lines are affixed to a single backbone rope, which is suspended between large buoys at each end and then anchored to the seabed. The vertical dropper lines provide the base for the mussels to attach and grow, allowing for easy management and inspection throughout their lifecycle [2].

As mussels begin to grow from the seeded droppers, the increased mass introduces significant drag forces which influence the movement of water within the farm. Research shows that suspended shellfish structures can reduce water currents and generate wakes that can change circulation patterns. It has been shown that mussel droppers change the flow behaviour by introducing resistance to currents both inside and around these shellfish farms [3]. These resistive effects influence the nutrient transport, biological productivity, and the sedimentation process of the seabed below. The impacts also extend beyond simply the environment, as current and flow changes can affect the stability of underwater vehicles. As seafood farming operations grow, automated monitoring technologies are being considered more heavily. Robotic systems capable of operating within this complex environment could provide improved monitoring of mussel growth and farm infrastructure while reducing the reliance and expense of current manual inspection methods.

2.2 Use of ROVs for Mussel Farm Inspection

ROVs have been widely established as a tool for inspections and monitoring in underwater environments. The ROV platform allows for a wide variety of data collection without the need for human divers or large vessel infrastructure. This reduces risks and costs for operators in these areas while also allowing for more frequent data collection and a higher level of consistency of the collected data. ROVs are particularly well-suited to inspection tasks because they can hover in place, carrying sensors such as cameras, sonar systems, and manipulators [4]. At present, these underwater ROVs are mostly used by offshore industries to inspect underwater infrastructure such as pipelines, cables, and structural components. Similar technologies are being explored by the seafood sector. In mussel farming operations, inspections are traditionally performed by either divers or boat-mounted cranes, which allows for a visual assessment of droplines and the collection of samples to monitor growth and health. This is expensive, labour-intensive and limited by weather conditions, water depth, and diver safety constraints.

Utilising ROVs for inspections and sampling offers the potential to automate these tasks. Underwater ROVs equipped with cameras and computer vision systems can assess mussel density, growth rates, and environmental conditions while simultaneously reducing operational costs and human risk. However, these robots must operate near the dropper lines whilst capturing clear imagery for analysis. This creates its own challenges, as environmental disturbances such as water currents, turbidity, and aquatic life can all impact image quality. Effective stabilisation capabilities are a key requirement for underwater ROVs.

2.3 Stabilisation and Control of Underwater Robots

The complex dynamic fluid flows underwater make the autonomous motion of an ROV highly challenging. Forces such as drag, buoyancy, and applied current forces all impact the motion of the robot in all six axes. These impacts are typically non-linear and thus difficult to predict and combat

without complex analysis of the environment. Thus, accurate motion control requires robust control systems capable of compensating for these forces while maintaining desired positions or trajectories [5].

Many ROVs rely on closed-loop control systems such as proportional integral derivative (PID) controllers because of their simplicity and reliability. However, more advanced control systems have been developed to improve control performance in dynamic environments. These include sliding mode control, adaptive control, and model predictive control; all have been investigated to enhance the rejection of disturbances and track the trajectory of the ROV [6, 7]. These systems use feedback from onboard sensors, including inertial measurement units (IMUs), depth sensors, and acoustic or light positioning systems such as sonar or lidar. Additionally, modelling these complex systems is an important part of underwater ROV control design. Mathematical models describing the interaction between the vehicle and the surrounding fluid allow engineers to predict how the ROV will respond to environmental influences such as currents. These models allow simulations to be performed to rapidly prototype and refine control systems for these underwater ROVs so they can move effectively underwater despite changing system conditions.

2.4 Modelling Fluid Motion in Ocean Structures

Modelling fluid motion in structured environments such as mussel farms is essential for understanding the forces acting on underwater vehicles. Unlike open-water conditions, mussel farms are a dense, complicated environment of dropper lines that interact with background currents, creating complex flows. These interactions generate turbulence, wakes, and variations in flow velocity, resulting in a dynamic, non-uniform system [3]. Such conditions can introduce unpredictable disturbances that affect the stability and control of ROVs.

In other studies, a range of modelling approaches have been developed to represent these flow behaviours. Simplified drag-based models are commonly used to approximate the resistive effects of droplines using drag coefficients [8]. These models are efficient and suitable for large-scale simulations but lack accuracy with local turbulence. More advanced methods, such as computational fluid dynamics (CFD), can provide detailed representations of flow interactions and wake structures within mussel farms. However, these approaches are often computationally expensive and less practical for real-time applications. For ROV stabilisation, the model must balance accuracy and computational efficiency. Reduced-order models capturing the dominant flows and frequencies can be used to predict environmental forces acting on the vehicle. This will allow for an improved control system to be developed. Thus, improving the performance of the ROV at stabilising against dropper lines.

2.5 Vision- and Sonar-Based Motion Estimation Underwater

Visual and acoustic sensing has been used to estimate motion underwater for some time. Feature-tracking approaches applied to underwater image sequences have been used to recover the motion of a vehicle relative to its surroundings [9], and similar feature-based tracking has been extended to sonar video for 3D motion estimation [10]. Sonar tracking has likewise been applied to real-time obstacle avoidance and path planning for underwater vehicles [11]. This shows that low-level tracked-feature data can directly support real-time robotic decision-making without requiring a full physics-based model of the surrounding environment.

More recently, site-specific hydrodynamic characterisation of mussel farms has become a more investigated area. A field study used in-situ measurements to assess the impact of an offshore longline mussel farm on local water circulation [12]. However, this and comparable studies, are oriented toward environmental and ecological outcomes (e.g. nutrient transport, farm siting) rather than toward producing a control-relevant, real-time representation of structure motion suitable for feeding a robot's stabilisation system.

3 Method

3.1 Research Objectives

The purpose of this proposal is to investigate and model fluid motion in ocean structures such as mussel farms. This is being done as a tangential research project to an underwater robot project, for which the findings of this research will be implemented as a stabilisation solution. This research project will aim to determine the feasibility of modelling these complex fluid systems, and if determined to be feasible, develop a custom model for use in the underwater ROV. The model will need to be provably accurate and computationally lightweight. If either of these objectives is not met, the model would not be suitable for implementation in the ROV.

In this context provable accuracy is defined by the operation and output of the prediction system. Comparing the predicted position of the mussel against its recorded position at the time of prediction, a level of accuracy can be found, primarily using the root-mean-square error (RMSE) as the means of quantification. This can then be compared against a version of the prediction algorithm that simply maintains the current position of the mussel, apply no prediction methods. This will act as a baseline for comparisons to be draw against. A viable model should be consistently more accurate than the holding baseline across a variety of test cases.

3.2 Approach

In order to achieve the aforementioned goals, the following workstreams will be undertaken.

3.2.1 Research into Preexisting Solutions

An initial investigation into existing approaches will be undertaken. This will draw on the mussel farm hydrodynamics and underwater-vehicle stabilisation literature reviewed in Section 2, as well as on how motion or position data for underwater structures has been captured and used elsewhere, to identify potential techniques usable for producing a lightweight, real-time-capable prediction model. Given the intended use of the ROV's own stereo camera tracking data as the input to this model, this investigation is likely to include signal-processing and state-estimation techniques capable of forecasting a noisy, discretely sampled position signal [13]. As such, methods such as frequency-domain (Fourier transform) extrapolation [14] and recursive filtering will be investigated for suitability.

3.2.2 Initial Implementations and Simulation

Using tracked mussel position data collected via the ROV's stereo camera, initial prediction models will be developed and tested in software, drawing on the candidate techniques identified in Section 3.2.1. Accuracy will be assessed by comparing each model's predicted positions against subsequently observed real positions, benchmarked against a simple baseline (Section 3.1). Models will be iterated on and refined until a method meeting the accuracy objective in Section 3.1 is identified, or until it is determined that no candidate method can meet them with the data available.

3.2.3 Refinements and Real-World Testing

Once an initial model has been developed that satisfies the criteria for implementation, it will be translated into a form suitable for use within a control system on the ROV. This will allow for real-world testing and further refinement of the model. While implementing the model on the physical ROV is not necessarily a requirement for this research project, doing so would provide a valuable additional check on the model's accuracy and computational lightness against the objectives in Section 3.1 under real operating conditions.

4 Results

4.1 Prediction Pipeline Methods

After conducting the investigation into potential prediction methods as per Section 3.2.1, three main techniques were for testing and potential implementation. The three methods included recursive Fast Fourier Transform (FFT) on a buffer filled from previous data, linear/polynomial interpolation on a buffer of previous data, and the Kalman filter family, specifically, the Unscented Kalman Filter [15]. These methods were to be implemented using python [16] and tested using a mixture of real-world data gathered from the ROV and synthetic data.

4.2 Method Verification

To test the accuracy of the chosen methods, repeated tests were run using four sets of real-world data. Results from these tests are shown below. Accuracy is measured as per Section 3.1 with the RMSE, and deviation in accuracy from a zero-order hold (ZOH) method acting as a baseline.

Table 1: Results from testing of prediction method accuracy. Tested over 4 data set. Buffer length of 10 data points and a prediction window of 5s.

Prediction Method	Metric	FFT	1st Order Interpolation	2nd Order Interpolation	UKF
Test Data 1	**RMSE (m)**	0.3827	0.3258	0.8907	0.1404
	% from ZOH	-172.0	-131.6	-533.2	-6.1
Test Data 2	**RMSE (m)**	0.2861	0.0214	0.0905	0.0155
	% from ZOH	-1561.0	-24.5	-425.6	+9.2
Test Data 3	**RMSE (m)**	0.3205	0.0955	0.6355	0.0188
	% from ZOH	-449.1	-63.6	-988.8	+67.9
Test Data 4	**RMSE (m)**	0.2314	0.1469	0.6474	0.1771
	% from ZOH	-320.5	-167.0	-1076.5	-52.7
Average	**RMSE (m)**	0.3052	0.1474	0.5661	0.0863
	% from ZOH	-625.7	-96.7	-756.1	+4.6

These results can be seen graphically in figure 2.

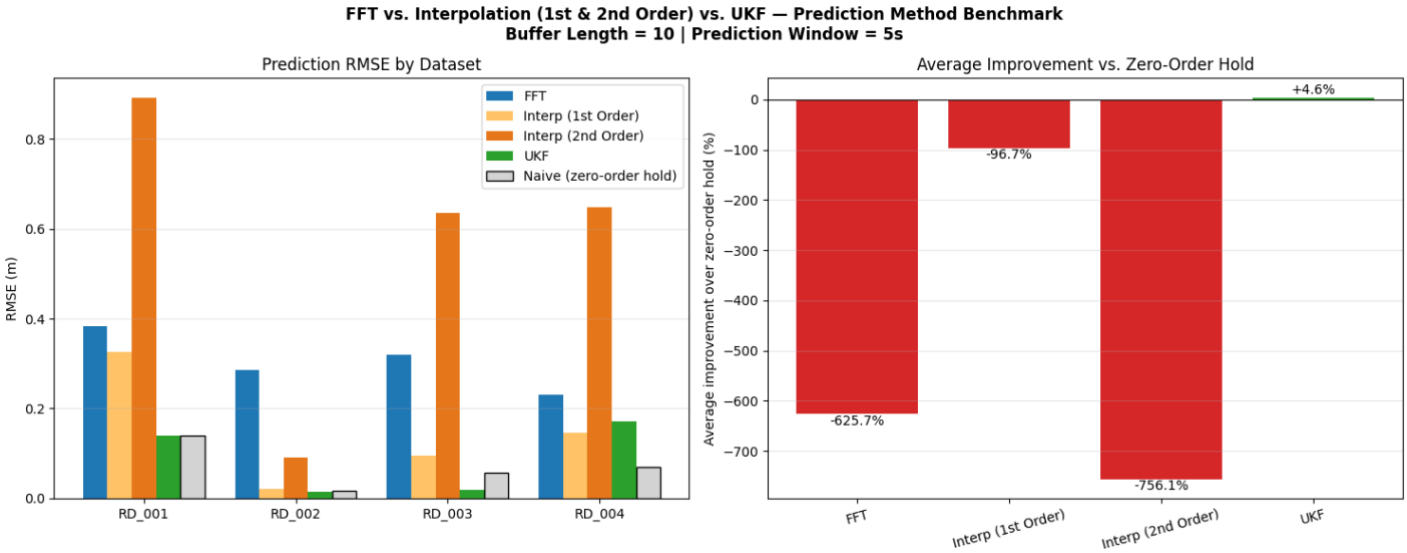


Figure 2: Graphical results from method verification testing.

5 Discussion

5.1 Findings

The FFT was the least reliable under test with an a average RMSE of 0.31m across the four tests that were ran. Given the scale of the system an error of 0.31m was deemed far too high. Compared to the zero-order hold the FFT method was found to be -653% less accurate. Investigation showed that while attenuation of the higher frequencies did yield a more smooth output from the prediction algorithm, it was too slow to respond to quick changes in position as these would be short “bursts” of high frequencies and would be filtered out, but the inclusion of these higher frequencies meant more sporadic prediction response. Overall the FFT method was not furthered due to its inherent inaccuracy.

The interpolation method proved to be better than the FFT. With an average RMSE of 0.15m and a reduction in accuracy compared to the zero-order hold of -97% using a first order interpolation implemented by the NumPy polyfit library. This is notably better than the FFT, but still not accurate enough for implementation on the ROV. Investigation suggested that over shorter prediction windows (<1s) it was fairly accurate, but as the prediction window grows, the accuracy recedes rapidly. This is especially apparent in instances where tracking was lost, which is a common occurrence due to the tracking method. Often when tracking is lost there is a spike in motion which causes the tracking algorithm to lose its position, this would cause the linear interpolation to rapidly overshoot as there would be no feedback to rectify until tracking was regained. Testing was also conducted with a second order interpolation, however this preformed notable worse with an RMSE of 0.57m and an accuracy regression of -751% compared to the zero order hold. This was found to be due to similar reasons as the linear method, being accentuated due to the polynomial attempting to fit to the sharp changes in position, causing more drastic overshoot. The interpolation method was also not furthered due to its inaccuracy.

The final method tested was the Unscented Kalman Filter (UKF). The UKF was chosen over the regular Kalman filter as it more reliably handles non-linear data [17]. The UKF was far more accurate than either of the previous methods with an RMSE of 0.086m and an improvement over the zero-order hold of 5%. Which while small, is a good sign. The additional benefit to the use of the UKF is that multiple instances can be run in parallel and they do not require buffer arrays, reducing the computational overhead of the prediction system. The UKF was chosen to the primary mechanism for prediction.

5.2 Implications

These results indicate that a vision-based approach using a UKF as a prediction method is a feasible solution to the proposed problem. While at the current stage of research and development it is not fully workable, yielding only a 4.6% increase over a zero-order hold approach, it still shows promise. Further improvements to the accuracy could result in more successful grab attempts of mussels on the dropper lines.

5.3 Limitations

There are several limitations to the results presented thus far, with only four sample sets tested the results are not overly comprehensive as of yet. However, work is being made on accumulating more data sets for additional testing and verification. Furthermore, while RMSE was chosen as a key metric for testing, it is worth noting that the RMSE may be inflated in instances where the tracking of the mussel is lost, which returns a position of (0,0,0). This means, while largely meaningful, in test cases where there is more pronounced loss of tracking, results will be more skewed to suggest inaccuracy of the model.

5.4 Future work

Future work that is currently being assessed for this project includes the following

- Coalesce data from multiple tracked mussels into a unified output.
- Investigate and test dynamically varying the Q and R matrices of the UKF to tune response based on live behaviour [18].
- Verify the constant-velocity approach taken by the UKF as opposed to oscillatory motion.
- Optimise the software for speed and minimal computational load when deployed on the ROV.
- Time permitting, live testing on the ROV.

An updated Gantt chart for this project has been included in Appendix A.

6 Conclusions

At this stage the research suggests that there is a feasible pathway to utilise computer vision tracking and subsequent motion prediction to enhance the grabbing accuracy of an autonomous mussel sampling ROV. The implemented UKF method does perform better than a non-responsive position holding baseline and there is room to improve upon this through the remainder of the project. As of now no concrete conclusions have been drawn but as mentioned above, research and testing suggest a positive outcome for this project.

7 Declarations

Author Contributions:

- **Alexander Pirie**: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Visualization, Writing – original draft, Writing – review & editing
- **Phuong Nguyen**: Data curation
- **Prof Richard Green, Tim Rensen**: Project administration.

Funding: N/A

Conflict of Interest: N/A

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AI usage: ChatGPT and Claude were utilised to draft the literature review talking points and aide with debugging testing software.

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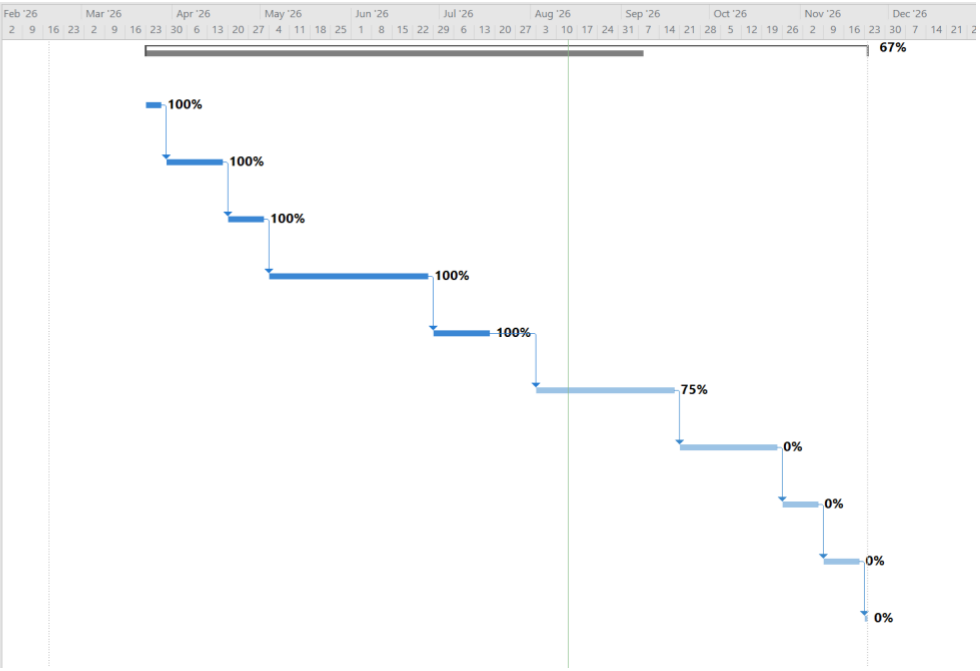
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Appendix A: Project Planning

	WBS	Task Mode	Task Name	Duration	Start	Finish	Cost	Resource Names
1	**1**		**Mussel Tracking and Position Prediction**	**176 days**	**Mon 23/03/26**	**Mon 23/11/26**	**$0.00**	**Alex**
2	1.1		Communicate with project technicians to gather current data and performance	5 days	Mon 23/03/26	Fri 27/03/26	$0.00	Alex
3	1.2		Investigate possible solutions	15 days	Mon 30/03/26	Fri 17/04/26	$0.00	Alex
4	1.3		Establish performance of existing systems in simulation software	10 days	Mon 20/04/26	Fri 1/05/26	$0.00	Alex
5	1.4		Develop initial model	40 days	Mon 4/05/26	Fri 26/06/26	$0.00	Alex
6	1.5		Test initial model with Real World Data	15 days	Mon 29/06/26	Fri 17/07/26	$0.00	Alex
7	1.6		Iterate upon model until suitable accuracy achieved	35 days	Mon 3/08/26	Fri 18/09/26	$0.00	Alex
8	1.7		Implement Method onto ROV for Testing of Efficacy	25 days	Mon 21/09/26	Fri 23/10/26	$0.00	Alex
9	1.8		Real world testing and further iteration	10 days	Mon 26/10/26	Fri 6/11/26	$0.00	Alex
10	1.9		Formalise findings	10 days	Mon 9/11/26	Fri 20/11/26	$0.00	Alex
11	1.10		Present solution and findings.	1 day	Mon 23/11/26	Mon 23/11/26	$0.00	Alex



Appendix B: H&S Hazard Assessment

Note: The Risk matrix identifies risks involved with the operation of the ROV in later testing.

Risks of system in its CURRENT STATE, with its existing controls.								Risks of system in its FUTURE STATE after these treatments. Insert any new threats caused by the treatments.			
Architecture level: Work-stream, project phase, hardware category, workstation	Specific hazard	Consequence (C), as per 'Severity of harm' scale 1. Hazard occurred 2. Incident with no harm 3. Incident and Minor harm 4. Incident and exposure to serious harm 5. Serious harm Occurs 8. Death 10. Catastrophe	Likelihood (L) of that harm consequence arising 6 Almost certain 5 Likely 4 Possible 3 Unlikely 2 Rare 1 Almost incredible	Risk (C x L)	Treatment? Consider Preventative and Recovery mechanisms. 30 or higher Unacceptable risk. 18 or higher Urgent treatment. 8 or higher Consider treatment 7 or less No intervention necessary.	Action required by who? Resources required?	Monitoring required of efficacy of treatment?	Treated Consequence (C*)	Treated Likelihood (L*)	Residual Risk (C* x L*)	Is this acceptable? What further action is required?
Sensors	Reads incorrect data	2	3	6	Minimal action required, code alterations	Alex	No	2	3	6	Yes - Further action not neccessary
Cameras	Electric fault	2	2	4	Ensure electrical rules followed when wiring	Alex	No	2	2	4	Yes - Further action not neccessary
Cameras	Water damage	2	2	4	Ensure ROV is closed fully and seals are intact	Alex	No	2	2	4	Yes - Further action not neccessary
Battery	Overheating	3	2	6	Check battery health/condition prior to operation and monitor power draw while in use	Alex	No	3	2	6	Yes - Further action not neccessary
Battery	Electric Fault	2	2	4	Ensure electrical rules followed when wiring	Alex	No	2	2	4	Yes - Further action not neccessary
Battery	Water damage	2	2	4	Ensure ROV is closed fully and seals are intact	Alex	No	2	2	4	Yes - Further action not neccessary
Propellers	Collision with object/ wildlife	3	4	12	Monitor surroundings, Ensure personal safety, before startup inspect propellers	Alex	Yes	3	3	9	Yes - Further action not neccessary
Propellers	Mechanical failure	3	2	6	Inspect propellors prior to use of ROV	Alex	No	3	2	6	Yes - Further action not neccessary
Motors	Electric fault	2	2	4	Ensure electrical rules followed when wiring	Alex	No	2	2	4	Yes - Further action not neccessary
Motors	Water damage	2	2	4	Ensure ROV is closed fully and seals are intact	Alex	No	2	2	4	Yes - Further action not neccessary
Motors	Overheating	3	2	6	Monitor power draw from motors while ROV in use	Alex	Yes	3	2	6	Yes - Further action not neccessary
Motors	Noise – environmental damage	2	4	8	Ensure motors don't result in propellor cavtation - monitor rpms	Alex	Yes	2	4	8	Yes - Further action not neccessary
LEDs	Electric fault	2	2	4	Ensure electrical rules followed when wiring	Alex	No	2	2	4	Yes - Further action not neccessary
LEDs	Water damage	2	2	4	Ensure ROV is closed fully and seals are intact	Alex	No	2	2	4	Yes - Further action not neccessary
LEDs	Bright lights	3	4	12	Proper PPE, Disable LED when it is out of the water	Alex	Yes	2	3	6	Yes - Further action not neccessary
Controllable Underwater Events	Collision – self inflicted	2	4	8	Montitor surrounding while operating and approach motion controll sensibly	Alex	Yes	2	4	8	Yes - Further action not neccessary
Uncontrollable Underwater Events	Strong current	2	3	6	Test ocean currents prior to operation and set current threshhold for non-operation	Alex	Yes	2	3	6	Yes - Further action not neccessary
Uncontrollable Underwater Events	Encounter with wildlife	4	4	16	Monitor surroundings and ensure no abundance of wildlife (fish, dolphins etc) in testing area	Alex	Yes	4	4	16	NO – not much we can do to mitigate this
Uncontrollable Underwater Events	Collision	2	3	6	Monitor surroundings and motion of robot while in operation and actively avoid collisions through piloting robot	Alex	Yes	2	3	6	Yes - Further action not neccessary